🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 6 · STANDARD 6.AT.6C

Write and Evaluate Numerical Expressions with Exponents

Lesson 6-4 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes

Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can write numerical expressions from a real situation and evaluate them using the order of operations, including powers.
Language Objective: I can explain my steps using the words expression, evaluate, order of operations, base, and exponent.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Numerical expression | Evaluate | Order of operations | Power | Base | Exponent

👆 *Tap any word to see its meaning and a picture.*

1. Numbers and operations with no equal sign and no variable form a ________ ________.

2. To put numbers in for the letters and work out the value is to ________.

3. Grouping symbols, then ________, then × and ÷, then + and −.

4. A number written with a base and an exponent, like 2^3, is a ________.

5. In 2^3, the number used as a factor is the ________.

6 The small raised number saying how many times the base is a factor is the

____________________ .

Write About the Math

How much will the ride cost? How does the expression model this situation?

¿Cuánto costará el viaje? ¿Cómo modela la expresión esta situación?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Numerical expression** *Expresión numérica*
- ☐ **Evaluate** *Evaluar*
- ☐ **Order of operations** *Orden de las operaciones*
- ☐ **Power** *Potencia*
- ☐ **Base** *Base*

Model: *Substitute the number of speakers for s, then evaluate.* — Students substitute s = 5 (150 + 12 × 5 = 210), explain 150 is the fixed venue fee and 12 is the cost per speaker, and note s can be any number.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The ride costs $____ because 2.50 + 1.75(____) = ____. The 2.50 is the ____ and 1.75 is the ____.**
- **My answer is ____ because ____.**

  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

  Mi afirmación es ____.
- **My evidence is ____, so ____.**

  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Numerical expression
2. **Notes 2:** Evaluate
3. **Notes 3:** Order of operations
4. **Notes 4:** Power
5. **Notes 5:** Base
6. **Notes 6:** Exponent
7. **Watch for this mistake:** *A common mistake with powers is multiplying the base by the exponent: reading 3^2 as 3 × 2 = 6 instead of 3 × 3 = 9. A second is evaluating strictly left to right, so 3 + 4 × 2 becomes 14 instead of 11. The order of operations exists so one expression has exactly one value: grouping symbols, then powers, then × and ÷ left to right, then + and − left to right.*
8. **Try It 1:** *— Grouping symbols outrank everything; powers outrank multiplying; multiplying outranks adding and subtracting.*
9. **Try It 2:** *— Each power is repeated multiplication of the base, as many times as the exponent says.*